

Yuta Hozumi

✉ yhozumi3@gatech.edu • Website Google Scholar

Current Position

Georgia Institute of Technology
Hale Visiting Assistant Professor
Mentor: Christine Heitsch

Atlanta, Georgia
2024–Present

Education

Michigan State University
Ph.D. Applied Mathematics
Advisor: Guo-Wei Wei

East Lansing, Michigan
2018–2024

Dissertation: *Topological representation and dimensionality reduction of single cell RNA sequencing and phylogenetic data*

Case Western Reserve University
BSc Applied Mathematics

Cleveland, Ohio
2014–2018

Research Interests

My primary research interests lies in geometric data analysis, with problems stemming from biology.

Courses Taught

Georgia Institute of Technology

MATH 3406, Second Course in Linear Algebra (small class)	Spring 2026; Fall 2026 (2 sections)
MATH 1554, Linear Algebra (large lecture)	Fall 2025 (2 sections)
MATH 1553, Introduction to Linear Algebra (large lecture)	Fall 2024
MATH 1552, Integral Calculus (large lecture)	Spring 2025

Michigan State University

MTH 124, Survey of Calculus (small class)	Summer 2019; Fall 2019; Spring 2020
---	-------------------------------------

Mentoring

Group Reading Project , Georgia Institute of Technology 2 students; final presentation on dimensionality reduction of large matrices.	Fall 2025
---	-----------

Group Reading Project , Georgia Institute of Technology 2 students; final presentation on machine learning for identifying cancer cells from gene expression data.	Fall 2024
--	-----------

Discovering America Research Project , Michigan State University 5 students; final presentation on deep learning models for analyzing cell trajectories from scRNA-seq data.	Spring 2024
--	-------------

Undergraduate Mentoring , Michigan State University Mentored Sean Cottrell on his honors thesis and collaborated on projects leading to 2 peer-reviewed publications. Now a Ph.D. student at the University of Georgia.	2023–2024
---	-----------

Publications

First Author Publications

8. **Yuta Hozumi**, Christine Heitsch, and Svetlana Poznanovic. Energy-Filtered Branching Alternatives Improve RNA Secondary Structure Recall. *Biophysical Journal* (2026).

7. **Yuta Hozumi**, and Guo-Wei Wei. Revealing the Shape of Genome Space via-mer Topology. *SIAM Journal on Life Sciences* 1, no. 2 (2026): 310-332.
6. **Yuta Hozumi**, Rui Wang, and Guo-Wei Wei. Correlated clustering and projection for dimensionality reduction. *Machine Learning: Science and Technology* 7, no. 3 (2026): 035062.
5. **Yuta Hozumi**, and Guo-Wei Wei. Analyzing scRNA-seq data by CCP-assisted UMAP and tSNE. *PloS One* 19.12 (2024): e0311791.
4. **Yuta Hozumi**, and Guo-Wei Wei. Analyzing single cell RNA sequencing with topological nonnegative matrix factorization. *Journal of Computational and Applied Mathematics* 445 (2024): 115842.
3. Sean Cottrell, **Yuta Hozumi**, and Guo-Wei Wei. K-nearest-neighbors induced topological PCA for single cell RNA-sequence data analysis. *Computers in biology and medicine* 175 (2024): 108497. (equally contributed)
2. **Yuta Hozumi**, Kiyoto Aramis Tanemura, and Guo-Wei Wei. Preprocessing of Single Cell RNA Sequencing Data Using Correlated Clustering and Projection. *Journal of Chemical Information and Modeling* (2023).
1. **Yuta Hozumi**, Rui Wang, Changchuan Yin, and Guo-Wei Wei. UMAP-assisted K-means clustering of large-scale SARS-CoV-2 mutation datasets. *Computers in biology and medicine* 131 (2021): 104264.

Co-Authored Publications

12. Faisal Suwayyid, **Yuta Hozumi**, Mushal Zia, JunJie Wee, Hongsong Feng and Guo-Wei Wei. CAKR: commutative algebra k-mer representations for genomics. *Nature Communications* (2026).
11. Mushal Zia, Faisal Suwayyid, **Yuta Hozumi**, JunJie Wee, Hongsong Feng, and Guo-Wei Wei. CAP: Commutative algebra prediction of protein-nucleic acid binding affinities. *Machine learning: science and technology* 6, no. 4 (2025): 045068.
10. Hongsong Feng, Faisal Suwayyid, Mushal Zia, JunJie Wee, Yuta Hozumi, Chun-Long Chen, and Guo-Wei Wei. CAML: Commutative Algebra Machine Learning. A Case Study on Protein-Ligand Binding Affinity Prediction. *Journal of Chemical Information and Modeling* 65, no. 13 (2025): 6732.
9. Hongsong Feng, Sean Cotrell, **Yuta Hozumi**, and Guo-Wei Wei Multiscale differential geometry learning of networks with applications to single-cell RNA sequencing data. *Computers in Biology and Medicine* 171 (2024): 108211.
8. Chen, Jiahui, Rui Wang, **Yuta Hozumi**, Gengzhuo Liu, Yuchi Qiu, Xiaoqi Wei, and Guo-Wei Wei. Emerging dominant SARS-CoV-2 variants. *Journal of Chemical Information and Modeling* 63, no. 1 (2022): 335-342.
7. Kaifu Gao, Rui Wang, Jiahui Chen, Limei Cheng, Jaclyn Frishcosy, **Yuta Huzumi**, Yuchi Qiu, Tom Schluckbier, Xiaoqi Wei, and Guo-Wei Wei. Methodology-centered review of molecular modeling, simulation, and prediction of SARS-CoV-2. *Chemical Reviews* 122, no. 13 (2022): 11287-11368.
6. Rui Wang, Jiahui Chen, **Yuta Hozumi**, Changchuan Yin, and Guo-Wei Wei. Emerging vaccine-breakthrough SARS-CoV-2 variants. *ACS infectious diseases* 8, no. 3 (2022): 546-556.
5. Rui Wang, Jiahui Chen, Kaifu Gao, **Yuta Hozumi**, Changchuan Yin, and Guo-Wei Wei. Analysis of SARS-CoV-2 mutations in the United States suggests presence of four substrains and novel variants. *Communications biology* 4, no. 1 (2021): 228.
4. Rui Wang, Jiahui Chen, **Yuta Hozumi**, Changchuan Yin, and Guo-Wei Wei. Decoding asymptomatic COVID-19 infection and transmission. *The journal of physical chemistry letters* 11, no. 23 (2020): 10007-10015.
3. Rui Wang, **Yuta Hozumi**, Yong-Hui Zheng, Changchuan Yin, and Guo-Wei Wei. Host immune response driving SARS-CoV-2 evolution. *Viruses* 12, no. 10 (2020): 1095.
2. Rui Wang, **Yuta Hozumi**, Changchuan Yin, and Guo-Wei Wei. Mutations on COVID-19 diagnostic targets. *Genomics* 112, no. 6 (2020): 5204-5213.
1. Rui Wang, **Yuta Hozumi**, Changchuan Yin, and Guo-Wei Wei. Decoding SARS-CoV-2 transmission and evolution and ramifications for COVID-19 diagnosis, vaccine, and medicine. *Journal of chemical information*

and modeling 60, no. 12 (2020): 5853-5865.

Awards and Scholarships

Travel Award, Great Lakes SIAM	September 2026
Travel Award, NSF-Simons National Institute for Theory and Mathematics	August 2025
College of Natural Science Completion Fellowship, Michigan State University	January 2023
Travel Award, Florida State University	October 2022
Herbert T. Graham Scholarship Award, Michigan State University	April 2020
SOURCE Summer Research Grant, Case Western Reserve University	Summer 2017
Research Education for Undergraduate, Case Western Reserve University	Summer 2016

Conferences Organized

Co-organizer with N. Hayes, JMM special session	2027
Co-organizer with M. Rana, AMS Southeastern Sectional special session	Fall 2026
Organizer, Southeastern Atlantic Section of SIAM minisymposium	Spring 2025
Organizer, SIAM Conference on Mathematics of Data Science minisymposium	Fall 2024
Organizer, SIAM Great Lakes Sectional Annual Meeting minisymposium	Fall 2023

Selected Talks

Joint Mathematics Meetings Special Session (Upcoming)	January 2027
SIAM Mathematics of Data Science Minisymposium (Upcoming)	November 2026
SIAM Texas-Louisiana Sectional Meeting Minisymposium (Upcoming)	October 2026
AMS Southeastern Sectional Special Session (Upcoming)	October 2026
North Carolina State University Biomath Seminar	September 2026
Midwest Mathematical Biology Seminar	September 2026
SIAM Great Lakes Sectional Minisymposium	September 2025
Southeastern Atlantic Section of SIAM Section Meeting Minisymposium	March 2025
SIAM Conference on Mathematics of Data Science Minisymposium	October 2024
Michigan State University Applied Mathematics Seminar	October 2023
Conference on Scientific Computing, Florida State University	October 2022
Calvin University Colloquium	February 2023
SIAM Great Lakes Sectional Minisymposium	October 2022
Michigan State University Computational Biology Forum	October 2021
Michigan State University Computational Biology Forum	October 2020

Professional Services

Outreach	
Directed Reading Program, Georgia Tech	Fall 2024, Fall 2025
Discovering America Research Project	Spring 2024
Journal Reviewer	
Nature Methods	

IEEE/ACM Transactions on Computational Biology and Bioinformatics
Biophysical Journal
Computational and Mathematical Bioinformatics and Biophysics
Computational and Mathematical Biomedical Engineering
Computer in biology and medicine
Journal of Chemical Information and Modeling